Hyperspectral Anomaly Detection via Structured Sparsity Plus Enhanced Low-Rankness

Yin-Ping Zhao, *Member, IEEE*, Hongyan Li, Yongyong Chen, *Member, IEEE*, Zhen Wang, *Senior Member, IEEE*, and Xuelong Li, *Fellow, IEEE*

***Abstract*— Hyperspectral anomaly detection (HAD), distinguishing anomalous pixels or subpixels from the background, has received increasing attention in recent years. Low-rank representation (LRR)-based methods have also been promoted rapidly for HAD, but they may encounter three challenges: 1) they adopted the nuclear norm as the convex approximation, yet a suboptimal solution of the rank function; 2) they overlook the structured spatial correlation of anomalous pixels; and 3) they fail to comprehensively explore the local structure details of the original background. To address these challenges, in this article, we proposed the structured sparsity plus enhanced low-rankness (S^2ELR) method for HAD. Specifically, our S^2ELR method adopts the weighted tensor Schatten-p norm, acting as an enhanced approximation of the rank function than the tensor nuclear norm (TNN), and the structured sparse norm to characterize the low-rank properties of the background and the sparsity of the abnormal pixels, respectively. To preserve the local structural details, the position-based Laplace regularizer is accompanied. An iterative algorithm is derived from the popular alternating direction methods of multipliers. Compared to the existing state-of-the-art HAD methods, the experimental results have demonstrated the superiority of our proposed S^2ELR method.**

***Index Terms*— Anomaly detection, Laplacian graph, low-rank, structure tensor, tensor decomposition.**

NOMENCLATURE

a	A scalar.
$\boldsymbol{a}$	A vector.
A	A matrix.
$\mathcal{A}$	A tensor.
$\boldsymbol{a}^T$	Transpose of $\boldsymbol{a}$.
A^T	Transpose of A.
$\mathcal{A}(:,:,k)$, $\mathcal{A}(:,k,:)$ and $\mathcal{A}(k,:,:)$	k-th frontal, lateral and horizontal slices of $\mathcal{A}$.
$\mathcal{A}_{i,j,k}$	(i,j,k)-th element of $\mathcal{A}$.
$\|\mathcal{A}\|_F = \sqrt{\sum_{ijk} \|\mathcal{A}(i,j,k)\|^2}$	Frobenius norm of $\mathcal{A}$.
$\hat{\mathcal{A}}_k = \text{fft}(\mathcal{A}, [], k),\ k = 1, 2, 3$	Discrete Fourier transform of $\mathcal{A}$ on mode-k.
$\mathcal{A} = \text{ifft}(\hat{\mathcal{A}}_k, [], k),\ k = 1, 2, 3$	Inverse discrete Fourier transform of $\mathcal{A}$ on mode-k.

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Yin-Ping Zhao is with the School of Software, Northwestern Polytechnical University, Xi'an 710072, China.

Hongyan Li is with the School of Mechanical Engineering, Northwestern Polytechnical University, Xi'an 710072, China.

Yongyong Chen is with the School of Computer Science and Technology and the Guangdong Provincial Key Laboratory of Novel Security Intelligence Technologies, Harbin Institute of Technology, Shenzhen 518055, China (e-mail: YongyongChen.cn@gmail.com).

Zhen Wang and Xuelong Li are with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China.

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) are 3-D images with two spatial dimensions and one spectral dimension, which have been used in a wide range of applications such as HSI classification [1], hyperspectral unmixing [2], object tracking [3], anomaly detection [4], etc. In the last few years, hyperspectral anomaly detection (HAD) has become a leading topic due to its increasing importance and attention in both military and agricultural matters. Normally, HAD refers to discovering anomalous pixels in HSIs by exploring the available spectral and spatial information.

Indeed, numerous promising AD and HAD methods have been presented, which can be roughly classified into three categories: *the classical Reed–Xiaoli (RX) methods based on statistical features, the representation-based methods, and the methods based on deep neural networks (DNNs)*. The representative model of the first category is the RX algorithm [5] by following the basic assumption that pixels of background obey the Gaussian distribution while that of the anomalies do not. Besides, Liu et al. [6] proposed two adaptive anomaly detectors based on statistical analysis to solve the anomaly detection problem with Gaussian noise. Unfortunately, the essential limitation of RX-based methods is that they are intractable to detect abnormal pixels when the background of HSIs being highly complex. To resolve the above problem, several enhanced variants of RX methods have been proposed such as the locally adaptable iterative RX (LAIRX) detector [7], the kernel-RX (KRX) [8], and the weighted-RXD [9]. These modified RX-based methods have improved the capability of handling complex anomalies to some degree. The second category of representation-based methods explored the spectral and spatial properties of HSIs by following the

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basic observation that the pixels in the background possess a strong correlation with their neighbors [10], [11], [12], being conceived as low rank, while anomalous pixels present rarely, being considered as sparse. For instance, Sun et al. [13] directly extended the low-rank and sparse matrix decomposition framework for AD (LRaSMD) and Zhang et al. [14] introduced an LRaSMD-based Mahalanobis distance method (LSMAD). However, they only focus on the global low-rankness of the background while ignoring the local sparsity [15]. To diminish this limitation, Feng et al. [16] proposed a tensor ring decomposition method coupled with a total variation regularization, which sufficiently exploited the conjunction between background and local anomaly pixels. Additionally, to reduce the impacts of abnormal pixels on the background representation, Hou et al. [17] proposed a background purification framework based on the linear representation. Unfortunately, over-constraining on the background would damage the smoothing among the background pixels. Inspired by Rudin et al. [18] and Osher et al. [19], the graph and total variation regularized low-rank representation (GTVLRR) method was investigated by Cheng and Wang [20] to guarantee the correlations on local pixels to produce a smooth background. And Zhao et al. [21] introduced an enhanced TV regularized representation model with an endmember background dictionary (ETVEBD) method to eliminate excessive destruction of the spatial structure. Nevertheless, the above matrix-based anomaly detectors fail to capture the 3-D structure of original HSIs, leading to a biased estimation of the background and misclassification of abnormal pixels. To optimally utilize the potential of 3-D tensor data, low-rank and tensor sparse decomposition (LRASTD)-based methods have recently gained considerable interest [22], [23], [24]. Qin et al. [25] modeled the background and anomaly as tensors and employed the tensor nuclear norm (TNN) to discipline the low-rankness of the background. Some other tensor technologies such as Tucker decomposition et al. [26], tensor-based principal component analysis (TPCA) [27] were used for HAD. Subsequently, the prior-based tensor approximation (PTA) method [28] was presented to better discover the low-rankness, which added the priors into both the spectral and spatial dimensions.

Meanwhile, the third category of DNN-based anomaly detectors has also drawn substantial attention. Li et al. [29] expanded the training data with the paired classification method within the limited dataset and identified anomalies by comparing the difference between the central and edge pixels in the HSI by the network. Then, Jiang et al. [30] employed an autoencoder (AE) network and an adversarial potential discriminator to reconstruct the desired images and combined a morphological filter with RX detector for AD. Fu et al. [31] fused a deep plug-and-play denoising regularization into the anomaly detector. However, most AD methods treat the background and anomalies separately and do not entirely utilize redundant spatial information. Hence, Zhao et al. [32] proposed a spectral–spatial hyperspectral AD method via collaborative representation constraint stacked AEs method to solve this issue. Afterward, Wang et al. [33] proposed a deep low-rank prior method combining the model-driven low-rank prior and the data-driven AE to achieve anomaly detection by exploiting the differences in reconstruction errors between background and anomalies based on the DNN.

For the first-category methods, the real-world HSIs' background may not always satisfy the Gaussian distribution, which drastically affects the abnormal detection performance. For the third-category methods, those DNN-based ones generally require lots of labeled data for training and face huge difficulty to choose suitable hyperparameters. Thus, in this article, we follow the second category. However, the second-category methods may suffer from some pending problems. On the one hand, the representation-based methods usually exploited the matrix nuclear norm and the TNN to separate the background and the anomaly of HSIs from their combinations such as [14], [27], [28], [34], and [35]. The nuclear norm is the convex approximation of the rank function, but unfortunately, the biased estimation since it treats all singular values equally without considering the different contributions of different singular values [36]. Consequently, there always remain inevitable errors in the estimation of the background. On the other hand, most existing HAD methods focus on the relations among the spectral dimensions while inherently ignoring the correlations in the spatial dimension, which is an extremely heartbreaking and sky-high loss. In addition, another essential characteristic of HSIs, i.e., the smoothness of the local structure of the background, is ignored by the current methods.

Thus, we aim to investigate one new hyperspectral anomaly detector, i.e., the structured sparsity plus enhanced low-rankness (S^2ELR) method, to address the above three limitations simultaneously. S^2ELR consists of three major components: the Schatten-p TNN, the structured spare norm, and the position-based graph Laplacian regularizer. Their basic motivations are as follows.

1) The tensor Schatten-p norm has achieved positive effects in object detection [37], tensor data recovery [38], multiview subspace clustering [39], etc. By adjusting the value of p, the tensor Schatten-p norm is the more flexible approximation function of the true rank function. For this reason, we are hopeful that it will also produce considerable benefits in HAD. To avoid deviating from the true rank caused by large singular values contributing too much to the rank, inspired by Wang et al. [40] and Shen et al. [41], we use the weighted tensor Schatten-p norm instead of the tensor Schatten-p norm. That is, we can achieve the rejection of large singular values by adjusting both the value of p and the weights to guarantee the proximity of the rank.
2) For abnormal pixels, we find that this bears strikingly similar to the extraction of subjects in the background separation field. Following [42], we introduce the structured sparsity-inducing norm to accomplish the detection of abnormal pixels. However, only 2-D data can be processed directly in [42], thus for 3-D HSIs, unfolding and descending are required. Additionally, we found that there are also some correlations of abnormal pixels

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between the adjacent frontal slices in HSIs. Therefore, we unfold the HSIs along the spatial dimension, and consequently, the abnormal pixels on the spatial dimension can also be considered to enhance the detection performance.

3) To alleviate the effect of noise in the original HSI and prevent the destruction of the local smoothness caused by the excessive constraints on the background, we introduce the position-based Graph Laplacian regularizer to protect the background.

In a nutshell, the main contributions of this article are summarized as follows.

1) We propose one novel HAD method without complicated dictionary construction, termed S^{2}ELR, which considers the globally low-rank property and the local smoothness of background and the structured sparsity of the anomaly simultaneously.
2) The weighted tensor Schatten-p norm is adopted to replace the TNN to better illustrate the low-rank characteristics of the background. Moreover, to preserve the local structure information, a position-based Laplacian is first introduced into S^{2}ELR.
3) To better separate anomalies from the background, we adopt an improved structured sparsity-inducing norm to depict abnormal pixels, which incorporates the abnormal pixels within each band on the spectral dimension. Experimental results also demonstrate that our method achieves superior performance compared to other existing state-of-the-art HAD methods.

The rest of this article is organized as follows. In Section II, we introduce some commonly used notations and definitions. Next, in Section III, we elaborate the proposed S^{2}ELR method in detail and present the optimization process. The experimental results and discussion are arranged in Section IV. Finally, the conclusion is given in Section V.

II. Notations and Definitions

A. Notations

The commonly used notations and their descriptions are presented in Nomenclature. For a third-order HSI tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times b}$, where n_1 and n_2 are the size of the spatial dimension and b is the size of the spectral dimension, we use $\boldsymbol{a}_{i,j} \in \mathbb{R}^{(1\times 1)\times b}$ to denote the pixel vector of the HSI $\mathcal{A}$ at position (i, j). $\hat{\mathcal{A}}_k$ denotes the result after the discrete Fourier transform (DFT) of tensor $\mathcal{A}$.

The notations of fold and its inverse unfold are defined as

$$A_k = \text{unfold}(\mathcal{A}, k) = \begin{pmatrix} A_k^{(1)} \\ A_k^{(2)} \\ \vdots \\ A_k^{(n_k)} \end{pmatrix}$$
$$\text{fold}(A_k, k) = \mathcal{A}, \ \ k = 1, 2, 3 \tag{1}$$

in which A_k and $A_k^{(i)}$ represent the matrix obtained by unfolding the tensor $\mathcal{A}$ and its ith slice along mode-k, respectively.

B. Definitions

In the following, some definitions and lemmas that will be involved in this article are presented.

Definition 1 (Matrix Nuclear Norm): For an arbitrary matrix $A \in \mathbb{R}^{n_1 \times n_2}$, where $A^T \in \mathbb{R}^{n_2 \times n_1}$ denotes its transpose, the matrix nuclear norm of A is defined as follows:

$$\|A\|_* = \text{tr}\left(\sqrt{A^T A}\right). \tag{2}$$

Definition 2 (Vector Infinite Norm): For an arbitrary n-dimensional vector $\boldsymbol{a} \in \mathbb{R}^n$, its infinite norm is defined as follows:

$$\|a\|_\infty = \max_{1 \leqslant i \leqslant n} |a_i|. \tag{3}$$

Definition 3 (Kronecker Product): For matrices $A = \{a_{ij}\}_{m\times n}$, $B = \{b_{ij}\}_{p\times q}$, the kronecker product between A and B is defined as follows:

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix}. \tag{4}$$

Definition 4 (TNN [43]): For an arbitrary third-order tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times b}$, the TNN is defined as follows:

$$\|\mathcal{A}\|_\circledast = \frac{1}{b} \sum_{i=1}^{b} \left\| A_k^{(i)} \right\|_* \tag{5}$$

where $\| \bullet \|_*$ denotes the matrix nuclear norm.

Definition 5 (Matrix Schatten-p Norm [44]): For an arbitrary matrix $A \in \mathbb{R}^{n_1 \times n_2}$, assume that $r = \min\{n_1, n_2\}$, $\{\sigma_i\}_{i=1,\ldots,r}$ is the singular values of A in descending order, then the Schatten-p norm can be defined as

$$\|A\|_{sp} = \left(\sigma_1^p + \sigma_1^p + \cdots + \sigma_r^p\right)^{1/p} \tag{6}$$

in which $0 < p \leqslant 1$ is an adjustable parameter.

It is evident that the matrix nuclear norm is a special case of the matrix Schatten-p norm when $p = 1$. Moreover, referring to the definition of TNN and matrix Schatten-p norm, we next present the definition of tensor Schatten-p norm.

Definition 6 (Tensor Schatten-p Norm [45]): For an arbitrary tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times b}$, the tensor Schatten-$p$ norm can be defined as

$$\|\mathcal{A}\|_{T_{Sp}} = \left(\frac{1}{b} \sum_{i=1}^{b} \left\| \hat{A}_3^{(i)} \right\|_{sp}^p \right)^{1/p} \tag{7}$$

where $\hat{A}_k^{(i)}$, $k = 1, 2, 3$ is the ith slice of $\hat{\mathcal{A}}$ on mode-k, $0 < p \leqslant 1$ is an adjustable parameter.

Definition 7 (Weighted Tensor Schatten-p Norm [46]): For an arbitrary tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times b}$, the weighted tensor Schatten-p norm can be defined as

$$\|\mathcal{A}\|_{w-T_{Sp}} = \left(\frac{1}{b} \sum_{i=1}^{b} W^{(i)} \left\| \hat{A}_3^{(i)} \right\|_{sp}^p \right)^{1/p} \tag{8}$$

in which $W^{(i)} = \text{diag}([w_1; w_2; \ldots; w_r])$, $r = \min\{n_1, n_2\}$.

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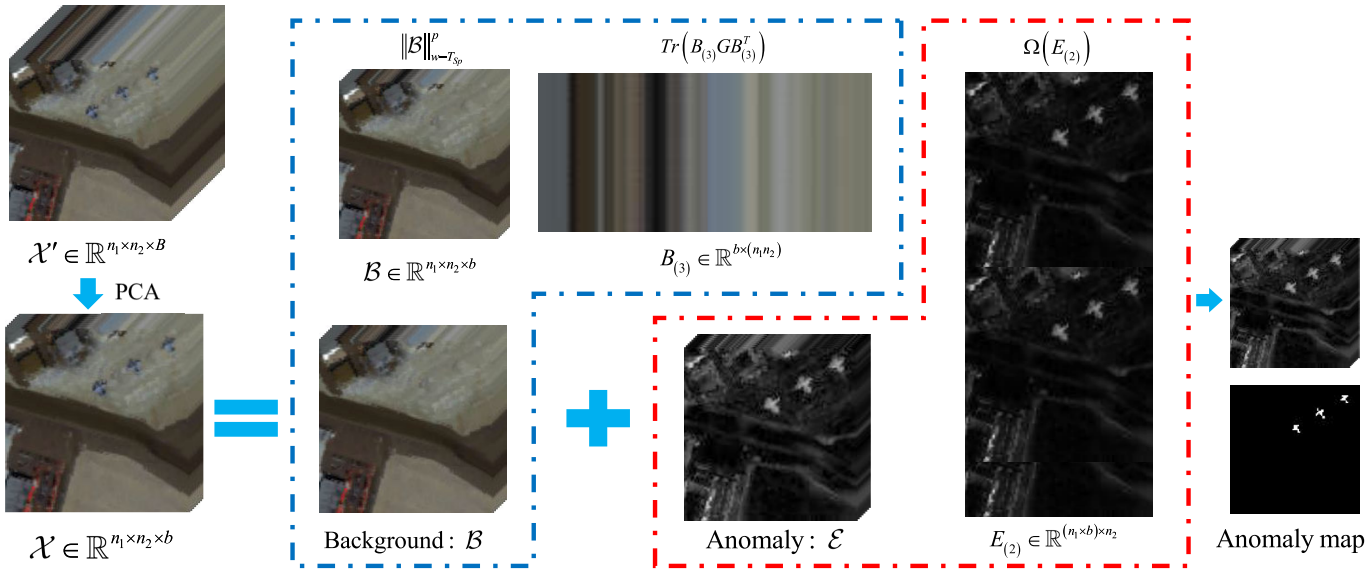


Fig. 1. Flowchart of the proposed S^{2}ELR method.

Then, the weighted tensor Schatten-p norm with power p of $\mathcal{A}$ can be expressed as

$$\|\mathcal{A}\|^p_{w-T_{Sp}} = \frac{1}{b}\sum_{i=1}^{b} \mathrm{tr}\left(W^{(i)} \hat{A}_3^{(i)p}\right) \tag{9}$$

where both $W^{(i)}$ and $\hat{A}_3^{(i)}$ are diagonal matrices.

Lemma 1 [47]: Assume that $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an arbitrary third-order tensor, $\hat{\mathcal{A}}_3 = \mathrm{fft}(\mathcal{A}, [], 3)$, then the following property is satisfied:

$$\|\mathcal{A}\|_F^2 = \frac{1}{n_3}\left\|\bar{A}_3\right\|_F^2 \tag{10}$$

where $\bar{A}_3 = b\mathrm{diag}(A_{(3)})$.

III. Proposed S^{2}ELR Method

The overall procedure of our proposed S^{2}ELR method is illustrated in Fig. 1. First of all, to diminish the computation complexity, we adopt the principal component analysis (PCA) into the original HSI for descending its dimension ($\mathcal{X}' \in \mathbb{R}^{n_1 \times n_2 \times B} \rightarrow \mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times b}$). After that, the dimension-reduced HSI is partitioned into two components: a low-rank background tensor and a sparse anomaly tensor. For the background, we employ a weighted tensor Schatten-p norm in conjunction with a position-based Laplacian regularization to constrain it, while a structured sparsity-inducing norm is adopted to characterize the anomaly. Finally, the anomaly map is obtained after the computation of the S^{2}ELR method.

A. Problem Statement

The purpose of AD is to separate the sparse abnormal pixels from the low-rank background. For a HSI tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times b}$, it can be decomposed into a low-rank background tensor $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times b}$ and a sparse anomaly tensor $\mathcal{E} \in \mathbb{R}^{n_1 \times n_2 \times b}$, represented as

$$\mathcal{X} = \mathcal{B} + \mathcal{E}. \tag{11}$$

Distinct from the matrix rank, the computation of tensor rank is highly complex. Since solving the rank minimization problem is NP-hard, the TNN, generalized from the matrix nuclear norm, is often utilized to characterize the tensor rank. However, TNN can lead to an inordinate contribution of large singular values to the rank, which adversely affects the estimation of tensor rank. To solve this problem, Hosono et al. [48] introduced the weighted TNN (WTNN), which penalizes small weights for large eigenvalues and large weights for small eigenvalues, thus changing the previous imbalance issue. Motivated by Shen et al. [41], the weighted tensor Schatten-p norm can be tuned to match the true tensor rank as closely as possible by adjusting the value of parameter p. Due to the constraint with the weight coefficients, it could be ensured that each singular value offers a uniform contribution to the rank. Consequently, the problem we need to resolve can be written as

$$\begin{aligned} &\min_{\mathcal{B},\mathcal{E}} \|\mathcal{B}\|^p_{w-T_{Sp}} + \mathrm{sparse}(\mathcal{E}) \\ &\text{s.t. } \mathcal{X} = \mathcal{B} + \mathcal{E}. \end{aligned} \tag{12}$$

Additionally, considering the interference of noise in the original HSI and avoiding the deterioration of local smoothness of the background caused by the undue low-rank constraints, we introduce an extra position-based Laplacian regularizer to restrict it. And thus, the model (12) can be rewritten as

$$\begin{aligned} &\min_{\mathcal{B},\mathcal{E}} \|\mathcal{B}\|^p_{w-T_{Sp}} + \mathrm{sparse}(\mathcal{E}) + \beta\mathrm{Graph}(\mathcal{B}) \\ &\text{s.t. } \mathcal{X} = \mathcal{B} + \mathcal{E} \end{aligned} \tag{13}$$

where Graph represents the Laplacian regularizer and β is a balance parameter. More specifically, the position-based

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Laplacian regularizer restricts the values around the pixel to be measured as close to it. In other words, given a radius r, with the measured pixel being the center of the circle, all pixels within its local range should share similar values. In this case, the remaining noise in the background can be removed, and the local smoothness can be also preserved.

Actually, it is challenging to obtain the position-based Laplacian regularizer. We unfold the background tensor $\mathcal{B}$ into a 2-D matrix $B_{(3)} \in \mathbb{R}^{b\times(n_1n_2)}$ along its spectral dimension. Subsequently, the mth pixel of $\mathcal{B}$ can be represented as $\mathcal{B}(i_m, j_m, :)$ or $\mathcal{B}_{(3)}(:, m)$, in which $m = (j_m - 1)n_1 + i_m$. Therefore, the measured mth pixel and its neighbors can be expressed as

$$U_{B_{(3)}} = \begin{bmatrix} m - kn_1 - k & m - k & m + kn_1 - k \\ m - kn_1 & m & m + kn_1 \\ m - kn_1 + k & m + k & m + kn_1 + k \end{bmatrix} \tag{14}$$

where $U_{\mathcal{B}_{(3)}} \in \mathbb{R}^{(2r+1)^2-1}$ indicates the set of neighbors for mth pixel, and $k = 1, 2, \ldots, r$ (r denotes the radius around the pixels to be measured). Finally, to bring the value of background pixels closer to the mth pixel, we introduce a weight matrix $W(m, n)$

$$W(m,n) = \begin{cases} e^{-(d_{mn}/\delta)}, & n \in U_{B_{(3)}} \\ 0, & n \notin U_{B_{(3)}} \end{cases} \tag{15}$$

where $d_{mn} = ((i_m - i_n)^2 - (j_m - j_n)^2)^{1/2}$ denotes the Euclidean distance between the mth and the nth pixels, and δ is the standard deviation of the values.

On the basis of (14) and (15), the position-based Laplacian regularizer can be constructed as

$$\begin{aligned} &\frac{1}{2}\sum_{m}^{(n_1n_2)}\sum_{n}^{(n_1n_2)} W(m,n)\left\|B_{(3)}(:,m) - B_{(3)}(:,n)\right\|_F^2 \\ &\quad = \mathrm{Tr}\left(B_{(3)}DB_{(3)}^T\right) - \mathrm{Tr}\left(B_{(3)}WB_{(3)}^T\right) \\ &\quad = \mathrm{Tr}\left(B_{(3)}GB_{(3)}^T\right) \end{aligned} \tag{16}$$

in which D is a diagonal matrix and its diagonal element is $D(m, m) = \sum_{n=1}^{n_1n_2} W(m, n)$. And then naturally, $G = D - W$ is the Laplacian matrix.

Thus, following [23], model (13) can be reformulated as:

$$\begin{aligned} &\min_{\mathcal{B},\mathcal{E}} \|\mathcal{B}\|_{w-T_{Sp}}^p + \mathrm{sparse}(\mathcal{E}) + \beta\mathrm{Tr}\left(B_{(3)}GB_{(3)}^T\right) \\ &\text{s.t. } \mathcal{X} = \mathcal{B} + \mathcal{E}, \quad B_{(3)} = \mathrm{unfold}(\mathcal{B}, 3). \end{aligned} \tag{17}$$

To further investigate their sparsity of the anomalous pixels, we unfold the original HSI along its spatial dimension and characterize the abnormal pixels with the structured sparsity-inducing norm, which is defined as

$$\Omega\left(E_{(2)}\right) = \sum_{l=1}^{n}\sum_{g\subset G}\left\|e_g^l\right\|_\infty \tag{18}$$

where $E_{(2)} = \mathrm{unfold}(\mathcal{E}, 2) \in \mathbb{R}^{(n_1b)\times n_2}$, $e_g^l \in \mathbb{R}^{(n_1b)}$ is the lth column vector of $E_{(2)}$ with sequences $\boldsymbol{l} = \{1, 2, \ldots, n_1b\}$, $g \subset G$ is a subset containing overlapping blocks of $\boldsymbol{l}$, and $\|*\|_\infty$ represents the infinite norm. In this article, to enhance the detection of abnormal pixels, distinct from [42] and [49], we define 2×2 overlapping-patch blocks, with each block duplicating 2 pixels adjacent to its neighbors.

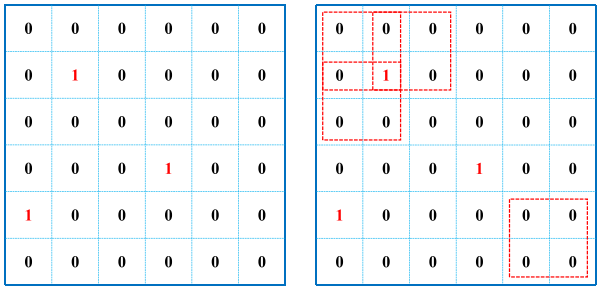


Fig. 2. Example of structured sparsity-inducing norm detection anomalies. (a) Example of a background with abnormal pixels. (b) Example of a sliding window to explore anomalies.

It is worth noting that the anomalies among each frontal slice are related and the continuity between anomalies and background also exist; hence, it is quite reasonable to use the structured sparsity-inducing norm unfolding along the spatial dimension to describe the abnormal pixels. Furthermore, an illustration is presented to facilitate an understanding of the process. As shown in Fig. 2, the background is denoted as 0 and the abnormal pixels are denoted as 1 and a 2×2 sliding block with two overlapping pixels is adopted to exploit the whole image step by step. Then, anomalies are identified by computing the maximum value in each sliding block. Indeed, to some extent, this is similar to the sliding window anomaly detector [50].

Finally, our proposed S^{2}ELR model is formulated as

$$\begin{aligned} &\min_{\mathcal{B},\mathcal{E}} \|\mathcal{B}\|_{w-T_{Sp}}^p + \alpha\Omega\left(E_{(2)}\right) + \beta\mathrm{Tr}\left(B_{(3)}GB_{(3)}^T\right) \\ &\text{s.t.} \begin{cases} \mathcal{X} = \mathcal{B} + \mathcal{E} \\ B_{(3)} = \mathrm{unfold}(\mathcal{B}, 3) \\ E_{(2)} = \mathrm{unfold}(\mathcal{E}, 2) \end{cases} \end{aligned} \tag{19}$$

where the first and third terms are the weighted tensor Schatten-p norm and the Laplacian regularizer, respectively, to discipline the background. The second term is the structured sparse-induced norm, which is adopted to sketch the abnormal pixels. Moreover, α and β are non-negative hyperparameters that balance the background structure and anomalous sparsity.

B. Optimization

To solve model (19), we first introduce an auxiliary variable $Z = B_{(3)}$, resulting in the following model:

$$\begin{aligned} &\min_{\mathcal{B},\mathcal{E},\mathcal{Z}} \|\mathcal{B}\|_{w-T_{Sp}}^p + \alpha\Omega\left(E_{(2)}\right) + \beta\mathrm{Tr}\left(ZGZ^T\right) \\ &\text{s.t.} \begin{cases} \mathcal{X} = \mathcal{B} + \mathcal{E}, \quad Z = B_{(3)} \\ B_{(3)} = \mathrm{unfold}(\mathcal{B}, 3) \\ E_{(2)} = \mathrm{unfold}(\mathcal{E}, 2). \end{cases} \end{aligned} \tag{20}$$

Then, we transform model (20) into the constraint-free optimization problem by providing its

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Algorithm 1 Solving for Weighted Tensor Schatten-p Norm Minimization

Input: $\mathcal{M} \in \mathbb{R}^{n_1 \times n_2 \times b}$, $n = \min\{n_1, n_2\}$.
Output: $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times b}$.

Compute $\hat{\mathcal{M}}_3 \in \text{fft}(\mathcal{M}, [], 3)$;
Compute SVD for each frontal slices $\hat{M}_3^{(\kappa)} = \hat{P}\hat{\Lambda}_\kappa \hat{Q}^H$, $\hat{\Lambda}_\kappa(i,i) = diag(v_1, \cdots, v_\tau)$, $1 \leqslant \kappa \leqslant b$, $1 \leqslant \tau \leqslant n$;
$\Theta w_{p,i,\kappa} = \frac{2-p}{2(1-p)} \times \left(2w_{i,\kappa}(1-p)\right)^{\frac{1}{2-p}}$;
for $i = 1 : n$ **do**
  for $\kappa = 1 : b$ **do**
    if $\hat{\Lambda}_\kappa(i,i) \leqslant \Theta w_{p,i,\kappa}$ **then**
      $\hat{\Lambda}_\kappa(i,i) = 0$;
    else if $\hat{\Lambda}_\kappa(i,i) > \Theta w_{p,i,\kappa}$ **then**
      fixed-point of the contraction:
      $\lambda \mapsto \hat{\Lambda}_\kappa(i,i) - w_{i,\kappa} p \lambda^{p-1}$;
  end
end
Compute $\hat{B}_3^{(\kappa)} = \hat{P}\left(\hat{\Lambda}_\kappa\right)\hat{Q}^H$;
Compute $\hat{\mathcal{B}}_3 = unfold\left(\hat{B}_3^{(\kappa)}, 3\right)$;
Return: $\mathcal{B} = \text{ifft}\left(\hat{\mathcal{B}}_3, [], 3\right)$.

augmented Lagrangian function

$$\begin{aligned}\mathcal{F}(\mathcal{B},\mathcal{E},\mathcal{Z},\mathcal{Y}_1,\mathcal{Y}_2) = {} & \|\mathcal{B}\|_{w-T_{Sp}}^p + \alpha\Omega\left(E_{(2)}\right) \\ & + \beta\text{Tr}\left(ZGZ^T\right) + \frac{\mu}{2}\left\|Z - B_{(3)} + \frac{\mathcal{Y}_1}{\mu}\right\|_F^2 \\ & + \frac{\mu}{2}\left\|\mathcal{X} - \mathcal{B} - \mathcal{E} + \frac{\mathcal{Y}_2}{\mu}\right\|_F^2 \end{aligned} \tag{21}$$

where $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are Lagrangian multipliers; μ is the penalty parameter. After this, we can solve problem (19) by dividing it into the following subproblems:

$$\begin{cases} \mathcal{B} = \underset{\mathcal{B}}{\text{argmin}} \|\mathcal{B}\|_{w-T_{Sp}}^p + \frac{\mu}{2}\left\|Z + \frac{\mathcal{Y}_1}{\mu} - B_{(3)}\right\|_F^2 \\ \qquad + \frac{\mu}{2}\left\|\mathcal{X} - \mathcal{B} - \mathcal{E} + \frac{\mathcal{Y}_2}{\mu}\right\|_F^2 & \text{(a)} \\ \mathcal{E} = \underset{\mathcal{E}}{\text{argmin}}\ \alpha\Omega\left(E_{(2)}\right) + \frac{\mu}{2}\left\|\mathcal{X} - \mathcal{B} - \mathcal{E} + \frac{\mathcal{Y}_2}{\mu}\right\|_F^2 & \text{(b)} \\ \mathcal{Z} = \underset{\mathcal{Z}}{\text{argmin}}\ \beta\text{Tr}\left(ZGZ^T\right) + \frac{\mu}{2}\left\|Z - B_{(3)} + \frac{\mathcal{Y}_1}{\mu}\right\|_F^2. & \text{(c)} \end{cases} \tag{22}$$

Next, we perform a separate solution for each subproblem by taking the alternating direction multiplier method (ADMM).

1) Update $\mathcal{B}$: For (22a), we simplify it into the following form:

$$\begin{cases} \mathcal{B} = \underset{\mathcal{B}}{\text{argmin}} \|\mathcal{B}\|_{w-T_{Sp}}^p + \frac{1}{2}\left\|\mathcal{B} - \mathcal{G}^{\mathcal{B}}\right\|_F^2 \\ \mathcal{G}^{\mathcal{B}} = \dfrac{\mu(\mathcal{X} - \mathcal{E} + \mathcal{Z}) + \mathcal{Y}_1 + \mathcal{Y}_2}{2\mu}. \end{cases} \tag{23}$$

Due to the fact that the weighted tensor Schatten-p norm is a typical nonconvex function, to facilitate its solution, we first consider the following problem:

$$\mathcal{B} = \underset{\mathcal{B}}{\arg\min}\ \mu\|\mathcal{B}\|_{w-T_{Sp}}^p + \frac{1}{2}\|\mathcal{M} - \mathcal{B}\|_F^2 \tag{24}$$

where $\mathcal{B}, \mathcal{M} \in \mathbb{R}^{n_1 \times n_2 \times b}$, μ is a balance parameter which is used to trade off the background fidelity with the low-rank term.

According to Lemma 1 and property (9), the minimization problem (24) can be derived into b subproblems as follows:

$$\hat{\mathcal{B}}_3 = \underset{\hat{\mathcal{B}}_3}{\arg\min} \sum_{i=1}^{b} \left(\left\|\hat{M}_3^{(i)} - \hat{B}_3^{(i)}\right\|_F^2 + \mu\left\|\hat{B}_3^{(i)}\right\|_{w-Sp}^p \right) \tag{25}$$

in which $\hat{B}_3^{(i)}$ denotes ith frontal slice of $\hat{\mathcal{B}}_3 = \text{fft}[\mathcal{B}, [], 3]$, and similarly, the same for $\hat{M}_3^{(i)}$. What's more, it is evident that $\hat{B}_3^{(i)}$ and $\hat{M}_3^{(i)}$ are complex matrices. To address problem (25), the following Lemma 2 is introduced.

Lemma 2 [51]: Suppose that M, $B \in \mathbb{C}^{n_1 \times n_2}$ are both complex matrices, and moreover, (25) is simply denoted as $\varphi(\hat{B}_3)$. And then, we define a mapping $\Phi : \mathbb{C}^{n_1 \times n_2} \to \mathbb{C}^{1 \times \kappa}$, which satisfied $\Phi(B) = \varphi \circ \vec{\sigma}_B = \varphi(\sigma_1, \sigma_2, \ldots, \sigma_\kappa)$, $\vec{\sigma}_B$ is the vector consisting of the singular values of B in a descending order, $1 \leqslant \kappa \leqslant n = \min\{n_1, n_2\}$. If $\Phi(B)$ meets the complex unitary invariance and $M = P\Lambda Q^H$ is the singular value decomposition (SVD) of M, we can obtain the optimum solution B^* for the following minimization:

$$\min_B\ \Phi(B) + \frac{1}{2}\|B - M\|_F^2 \tag{26}$$

where $B^* = P\Lambda_B^* Q^H$, and $\Lambda_B^* = \text{diag}(\vec{\sigma}_B^*)$ is the solution of $\min_{\vec{\sigma}} \varphi(\vec{\sigma}) + (1/2)\|\vec{\sigma} - \vec{\sigma}_M\|$.

Subsequently, by fixing other variables and using Lemma 2, the problem (23) can be solved by Algorithm 1.

2) Update $\mathcal{E}$: For (22b), we rewrite it as the following equivalent form:

$$\mathcal{E} = \underset{\mathcal{E}}{\text{argmin}}\ \alpha\Omega\left(E_{(2)}\right) + \frac{\mu}{2}\left\|E_{(2)} - G^{E_{(2)}}\right\|_F^2 \tag{27}$$

in which $\Omega(E_{(2)})$ indicates the structured sparsity-inducing norm. And $G^{E_{(2)}} = X_{(2)} - B_{(2)} + (Y_{2(2)}/\mu)$.

And thus, the problem (27) can be easily solved by the proximal operator for overlapping groups [49]

$$E_{(2)} = \text{prox}_g\left(G^{E_{(2)}}\right). \tag{28}$$

Accordingly, $\mathcal{E}$ can be yield as

$$\mathcal{E} = \text{fold}\left(E_{(2)}, 2\right). \tag{29}$$

3) Update $\mathcal{Z}$: For (22c), we take the derivative of Z and set its gradient to zero, then we can obtain

$$\beta ZG + \beta ZG^{\mathrm{T}} + \mu Z_{(3)} + Y_{1(3)} - \mu B_{(3)} = 0. \tag{30}$$

Furthermore, we get

$$Z = \left(\mu B_{(3)} - Y_{1(3)}\right)\left(\beta(G + G^T) + \mu I\right)^{-1}. \tag{31}$$

Finally, $\mathcal{Z}$ can be derived by

$$\mathcal{Z} = \text{fold}(Z, 3). \tag{32}$$

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Algorithm 2 The proposed S^{2}ELR for HAD

Input: The original HSI tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times b}$, p.
Output: The anomaly map $E \in \mathbb{R}^{n_1 \times n_2}$ and clean background $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times b}$.
Initialization: $\mathcal{B} = \mathcal{E} = 0$, $\mathcal{Y}_1 = \mathcal{Y}_2 = 0$, $\mu = 10^{-3}$, $\mu_{max} = 10^5$, $\rho = 1.1$, $\varepsilon_\Delta = 10^{-3}$, $\tau = 0$, $Iter_{max} = 150$;
while $\tau \leqslant Iter_{max}$ *or* $\varepsilon_{\max} < \varepsilon$ **do**
  Update $\mathcal{B}$ by Algorithm 1;
  Update $\mathcal{E}$ by Eq. (29);
  Update $\mathcal{Z}$ by Eq. (32);
  Check the convergence conditions: $\max(\Delta_\mathcal{B}, \Delta_\mathcal{E}, \Delta_\mathcal{Z}) \leqslant \varepsilon_\Delta$;
  Update the Lagrange multipliers $\mathcal{Y}_1$, $\mathcal{Y}_2$ and penalty parameter μ by Eq. (33);
end
Return: Anomaly map E and background $\mathcal{B}$.

4) Update $\mathcal{Y}_1$, $\mathcal{Y}_2$ and μ: The Lagrange multipliers and penalty parameter can be updated as follows:

$$\begin{cases} \mathcal{Y}_1 = \mathcal{Y}_1 + \mu \times (\mathcal{Z} - \mathcal{B}) \\ \mathcal{Y}_2 = \mathcal{Y}_2 + \mu \times (\mathcal{X} - \mathcal{B} - \mathcal{E}) \\ \mu = \min\{\rho \times \mu, \mu_{\max}\} \end{cases} \tag{33}$$

where $\mu_{\max}$ denotes the maximal value of μ.

Based on the above discussion, the entire solving procedure is summarized in Algorithm 2 and $\Delta_\mathcal{B}$, $\Delta_\mathcal{E}$, and $\Delta_\mathcal{Z}$ indicate the relative error after each update, respectively.

IV. Experiments and Discussions

A. Real Data Experiments

In this section, several real HSI datasets are utilized to test the performance of our algorithm, which are listed below.

1) Hyperspectral Digital Imagery Collection Experiment (HYDICE)-Urban Dataset [52]: It was acquired by the HYDICE sensor with the size of 80 × 100 pixels. Its spectral and spatial resolution are, respectively, 10 nm and 1 m, leaving 175 bands manipulation and analysis after removing noise and water absorption bands. Within this image, 21 abnormal pixels are presented, which consist of different vehicles and roofs in the background. The pseudo-color image (the eighth, 18th, and 37th bands) and the anomaly map are shown in the first row of Fig. 3(a) and (b), respectively.

2) San Diego Dataset [35]: It was collected by the airborne visible/infrared imaging spectrometer (AVRIS) sensor over the State of California, USA, with 100 × 100 pixels in spatial and 189 bands in spectral dimension. Within this view, three aircraft are considered as anomalies, while grass, roofs, and shadows belong to the background. Similarly, its pseudo-color image (the eighth, 18th, and 37th bands) and ground truth of the anomaly map are presented in the second row of Fig. 3(a) and (b).

3) Airport–Beach–Urban (ABU) Dataset[1]*:* It consists of four airports, four beaches, and five urban scenes. Except for beaches that are extracted by the reflective optical system imaging spectrometer (ROSIS), all of the others are obtained by AVRIS sensors [53]. Due to limitations of computing capacity, only four Airport datasets are displayed in Fig. 3, with the spatial size of 100 × 100 and spectral size of 205 (Airport-1–3) or 191 (Airport-4), while the remaining four Beach and five Urban datasets are shown in Fig. 4. Additionally, each dataset carries a manually labeled map of anomalies, which is illustrated in the last four rows of Fig. 3(b) and Fig. 4(b).

[1]http://xudongkang.weebly.com/

B. Detection Performance

We employ several quantitative indicators to quantitatively evaluate the performance of all methods. The first is the 3-D receiver-operating characteristic (ROC) curve that captures the variations between the detection probability PD and the false alarm rate PF [54]. And compared to the 2-D ROC curve, a threshold τ is inserted to measure the detectability of anomalies and the rejection of the background of the detector. Second, on the basis of 3-D ROC curve, we can obtain three different area under the ROC curve (AUC) values in different orientations, which are $\text{AUC}_{(\text{PD},\text{PF})}$, $\text{AUC}_{(\text{PD},\tau)}$, and $\text{AUC}_{(\text{PF},\tau)}$ under the 2-D ROC curve of (PD, PF), (PD, τ), and (PF, τ), respectively. Without loss of generality, larger $\text{AUC}_{(\text{PD},\text{PF})}$ and $\text{AUC}_{(\text{PD},\tau)}$ indicate a stronger extraction of anomalies and a smaller $\text{AUC}_{(\text{PF},\tau)}$ indicates a clearer suppression of background. Finally, following [55], we combine all three AUC metrics denoted as AUC_{OD} to evaluate the global performance, which can be computed as:

$$\text{AUC}_{\text{OD}} = \text{AUC}_{(\text{PD},\text{PF})} + \text{AUC}_{(\text{PD},\tau)} - \text{AUC}_{(\text{PF},\tau)}. \tag{34}$$

Obviously, a higher AUC_{OD} value means a superior exception detector performance.

To demonstrate the effectiveness of our proposed S^{2}ELR method, we compared it with the existing state-of-the-art methods, including: one classic method: global RX [5], two representation-based methods: RPCA [34] and TPCA [27], one GTVLRR method: GTVLRR [20], one low-rank representation (LRR) method: LRASR [35], one low-rank and sparse matrix decomposition method: LSMAD [14], one deep convolution neural network-based method (DeCNN-AD [31]), and one PTA method: PTA [28]. All methods are performed on MATLAB 2020a with Intel Core i7-12700K CPU at 3.60 GHz and 64 GB RAM.

1) HYDICE-Urban Dataset: The first row of Fig. 3 visualizes the detection maps of HYDICE-Urban scene obtained from all the tested methods. It can be observed that TPCA presents the worst visual appearance due to the mixing of anomalies and background pixels. Though the visual images of RX, RPCA, and GTVLRR are extremely similar, they vary considerably in detecting roofs over grass, where GTVLRR yields relatively pure and legible abnormal pixels. Subsequently, DeCNN-AD obtains favorable views on both vehicles and roofs. There is virtually no obvious visual distinction between the proposed S^{2}ELR and PTA, whereas PTA achieves equally sharper anomalous pixels. In addition, we sketch the 3-D ROC curves for these matching methods

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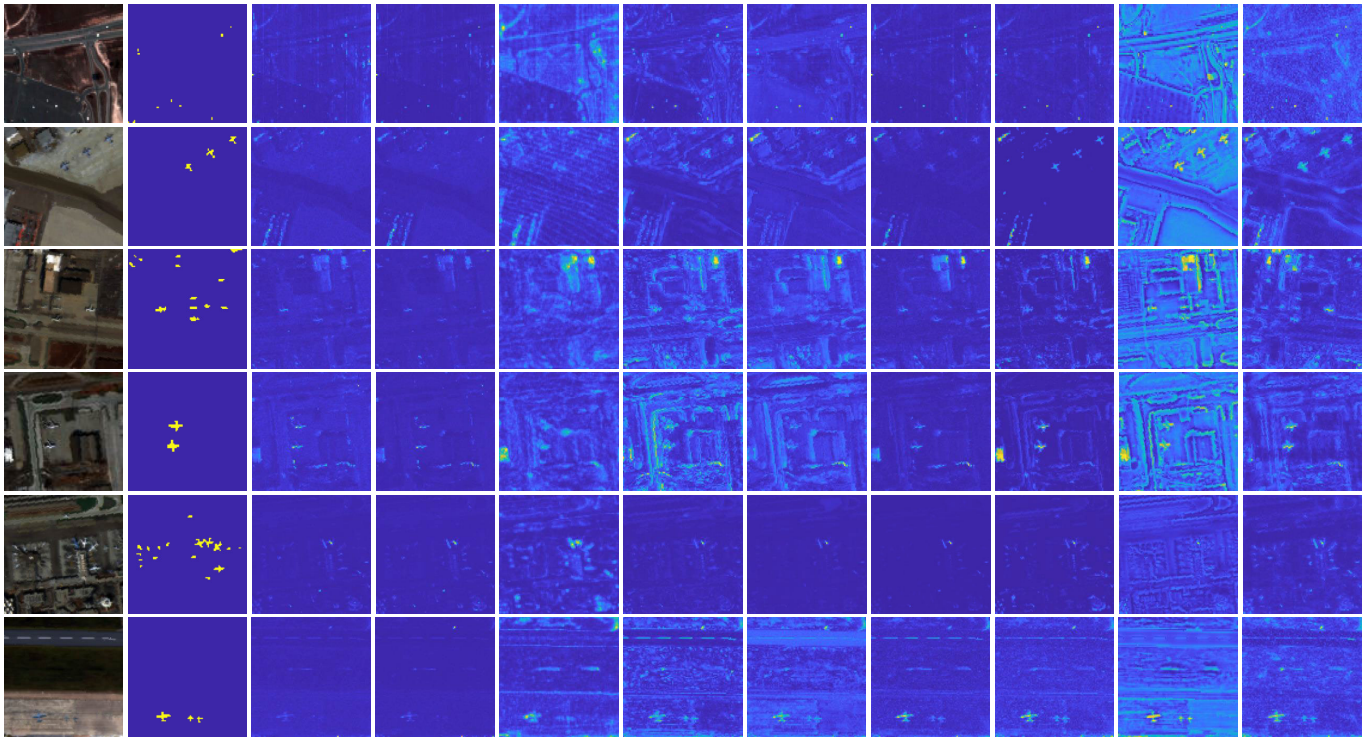


Fig. 3. Detection maps of the proposed S^{2}ELR and the compared methods for six datasets: (top to bottom) HYDICE-Urban, San Diego, and Airport 1–4. (a) Pseudo-color images of six scenes. (b) Ground truth. (c) RX [5]. (d) RPCA [34]. (e) TPCA [27]. (f) GTVLRR [20]. (g) LRASR [35]. (h) LSMAD [14]. (i) DeCNN-AD [31]. (j) PTA [28]. (k) S^{2}ELR.

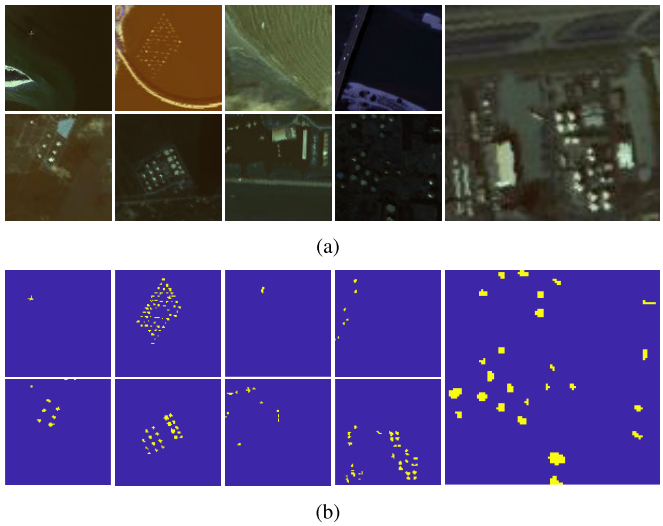


Fig. 4. (a) Pseudo-color images and (b) ground truth of anomalies for Beach and Urban in the ABU dataset. (Left to right, first row) Beach 1–4, respectively. (Left to right, second row) Urban 1–4 and the image on the far right is Urban 5.

in Fig. 5(a). Coherent to the visual results, TPCA shows the weakest performance. Though DeCNN-AD and GTVLRR share similar performance, GTVLRR has a lower PF value and therefore provides a better overall result. We also present the numerical evaluation metrics ($AUC_{(PD,PF)}$, $AUC_{(PD,\tau)}$, and $AUC_{(PF,\tau)}$) for all the methods mentioned above in Table I, with the optimal cases in bold and the suboptimal cases underlined.

2) San Diego Dataset: Similarly, the detection results of different methods on San Diego scene are illustrated in the second row of Fig. 3. The three aircrafts are clearly classified as abnormal pixels. However, RX, RPCA, and TPCA do not identify the anomalies in the image accurately, which is also supported by the metrics on the second row in Table I. Instead, LSMAD, LRASR and GTVLRR are increasingly visible, though only blurred anomalies can be acquired. Compared to PTA, S^{2}ELR can easily identify three aircrafts and receive a smoother background due to the position-based Laplacian regularization. Also, the 3-D ROC curves for all verified methods on San Diego dataset are displayed in Fig. 5(b). It is striking that due to the high PD value of the PTA in the earlier part of the curve, leading to the highest AUC_{OD} value. However, the PF value is also too expensive to allow a better visual effect than the proposed S^{2}ELR. Besides the optimal numerical result gained by PTA, the proposed S^{2}ELR and GTVLRR led to the suboptimal and third-optimal results, respectively.

3) ABU Dataset: We also exhibit the anomaly detection maps for the four Airport datasets in the last four rows of Fig. 3. As we can see, each image contains a different number of aircrafts acting as abnormal pixels. Compared to the other two Airport datasets, Airport-1 and Airport-3 datasets exhibit greater heterogeneity in both numbers and locations of aircrafts, as well as the proximity of adjacent aircrafts. This makes them more susceptible to low-rank misclassification, where aircrafts are incorrectly classified as background pixels. As a result, most methods do not perform well on these datasets. Fortunately, since S^{2}ELR exploits the details between

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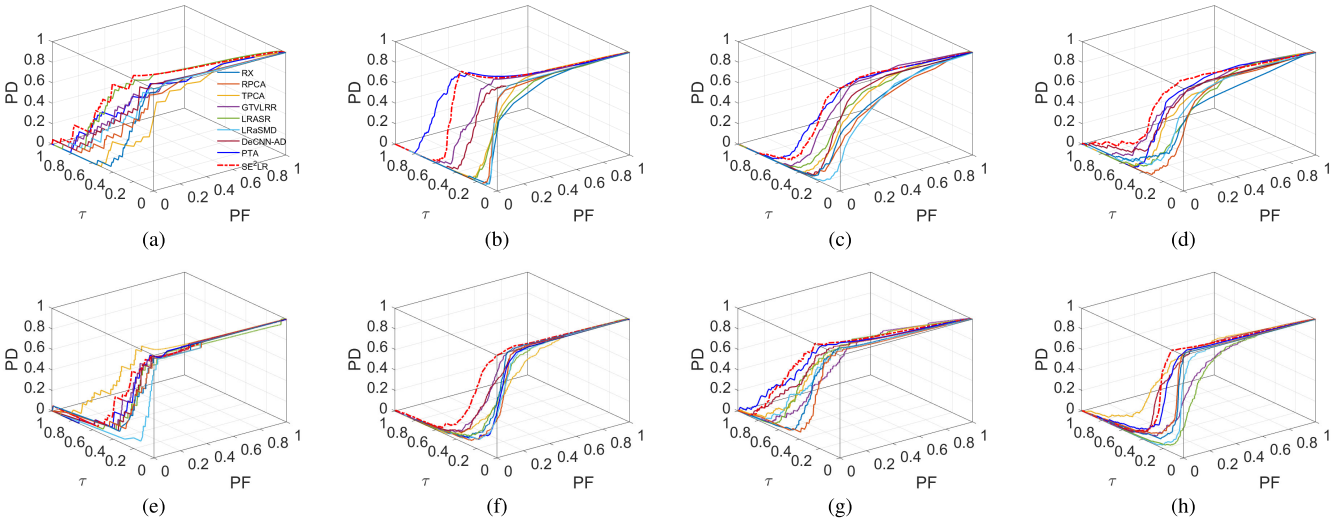


Fig. 5. ROC curves under various methods for HSI datasets. (a) HYDICE-Urban. (b) San Diego. (c) Airport-1. (d) Airport-2. (e) Beach-1. (f) Beach-2. (g) Urban-1. (h) Urban-2.

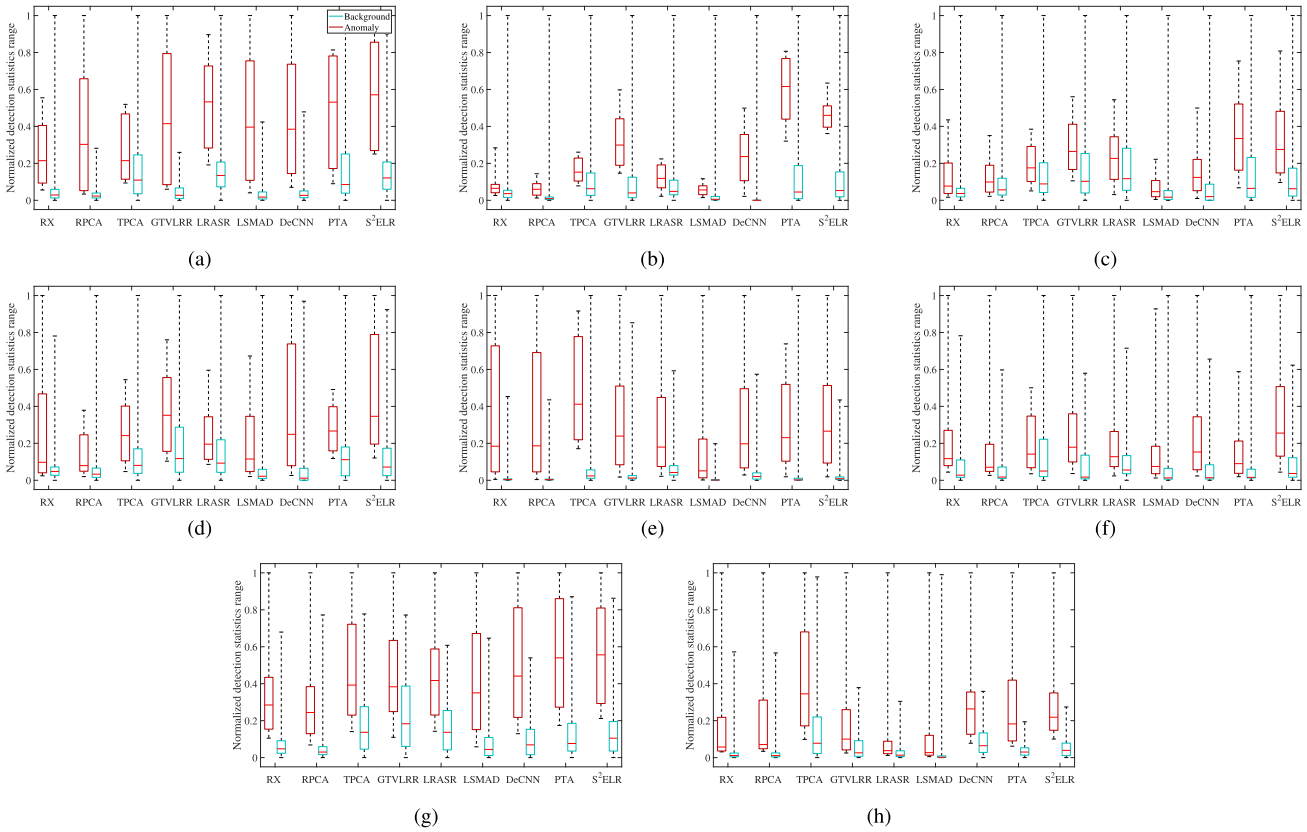


Fig. 6. Box plots of eight different datasets on all compared methods. (a) HYDICE-Urban. (b) San Diego. (c) Airport-1. (d) Airport-2. (e) Beach-1. (f) Beach-2. (g) Urban-1. (h) Urban-2.

each frontal slice of the original image, it is capable to discover additional anomalous pixels in combination with the improved structured sparsity-inducing norm. Furthermore, for Airport-2 and Beach-3 datasets, our S^{2}ELR method beats the suboptimal ones by nearly 20% and 10% higher for AUC_{OD} metrics in Table I, respectively. Nevertheless, the remaining comparison methods, such as RX, RPCA, TPCA, and LRASR, produce relatively poor results, again evidenced by the visual effects, which are insensitive to aircrafts stationed at the airport. Slightly different from the previous results, the proposed S^{2}ELR delivers suboptimal performance on a few datasets, such as Airport-4, Beach-1, and Urban-2. Specifically, it is striking to notice that in these datasets there are wide areas of pure dark background, which results in a sharp contrast between the background and the anomaly, and thus deteriorating the detection performance. Fortunately,

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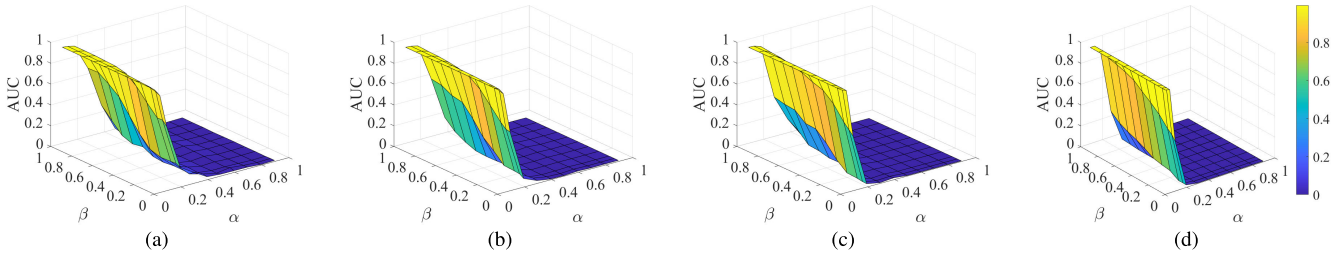


Fig. 7. AUC values for San Diego dataset under various p ($r = 5$). (a) $p = 0.1$. (b) $p = 0.3$. (c) $p = 0.5$. (d) $p = 0.7$.

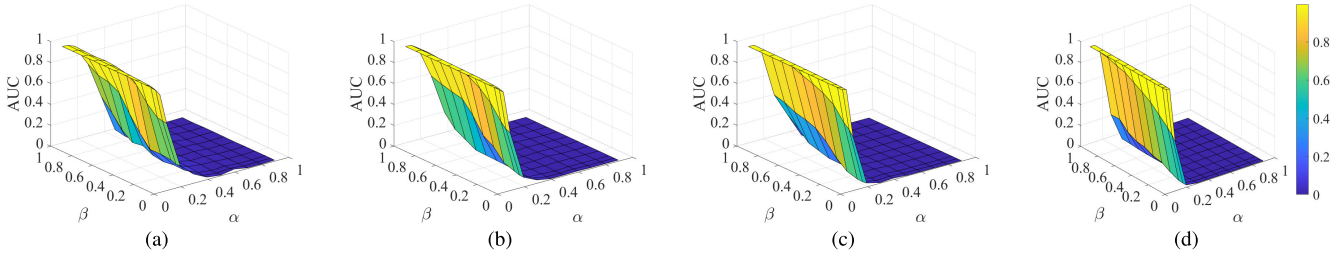


Fig. 8. AUC values for San Diego dataset under various p ($r = 9$). (a) $p = 0.1$. (b) $p = 0.3$. (c) $p = 0.5$. (d) $p = 0.7$.

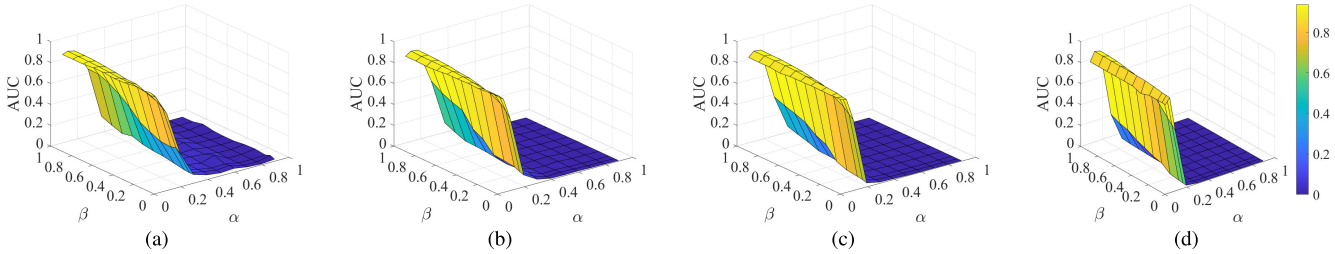


Fig. 9. AUC values for Airport-1 dataset under various p ($r = 5$). (a) $p = 0.1$. (b) $p = 0.3$. (c) $p = 0.5$. (d) $p = 0.7$.

it is still close to the best results and even better than the optimal values in some indicators. What's more, as shown in the last row of Fig. 3(j) and (k), by contrast with the optimal PTA, we only have a poorer detection in the first aircraft and we get a smoother background. Nevertheless, for the Urban-4 and Urban-5 datasets, as shown in Fig. 4, due to the high fusion of background and anomaly pixels, the anomaly is separated limited from the background, which is also extremely challenging DNN-based methods. The ROC curves are displayed in Fig. 5(c)–(h). As we can see, the curves for GTVLRR and LRASR are quite close to each other which also mirrors the numerical results in Table I. As well, RX and RPCA yield the worst results, with the visual results barely visible in the last row of Fig. 3(c) and (d). Moreover, in Fig. 5(g) and (h), relating to the numerical results in Table I, although our S^{2}ELR method is not optimal, there are still some indicators that achieve promising results.

Additionally, to better visualize the ability to separate the anomalies from the background, we have introduced box plots in Fig. 6. The green boxes are used to indicate the distribution of background pixels, while the red boxes are used to indicate the distribution of anomalous pixels. Further, the relative position between the green and red boxes represents the separability of the background and anomalies. To be more detailed, the larger distance between the two, the more effective the method of anomaly detection works. As can be observed in Fig. 6, the proposed S^{2}ELR method shows an excellent separation of anomalies from the background in most cases compared to RX, RPCA, LRASR, LSMAD, and DeCNN-AD. Whereas for the Beach-1 and Urban-2 datasets [see Fig. 6(e) and (h)], TPCA performs a little better than our S^{2}ELR, yet our method is still second-best, while for the other datasets, TPCA is less effective. On balance, our S^{2}ELR method yields a better separation of anomalies from the background than other compared methods.

C. Discussions

1) Effect of Different Parameters on Results: There are four parameters involved in our proposed method: the power p of the weighted tensor Schatten-p norm, the balance parameter α of the structured sparsity-inducing norm, β of the position-based Laplace regularizer, and the radius r of the neighboring pixels for constructing the Laplacian graph. Specifically, r is determined by the set $[3, 5, 7, 9, 11]$, α and β are taken as a range from 0.01 to 0.1 in steps of 0.01 in conjunction with 0.1 to 0.9 in steps of 0.1, while p is only taken from 0.1 to 0.9 in each interval of 0.1. As shown in

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TABLE I
AUC VALUES FOR ALL COMPARED METHODS

HSI Data Sets	AUC Metrics	RX	RPCA	TPCA	GTVLRR	LRASR	LSMAD	DeCNN-AD	PTA	S^2ELR
HYDICE-Urban	$AUC_{(PD,PF)}$	0.9857	0.9846	0.8242	0.9874	0.9882	0.9911	0.9976	0.9274	0.9946
	$AUC_{(PD,\tau)}$	0.2404	0.3368	0.2655	0.4695	0.5351	0.4028	0.4263	0.4828	0.5871
	$AUC_{(PF,\tau)}$	0.0351	0.0252	0.1296	0.0342	0.1415	0.0239	0.0309	0.1233	0.1298
	AUC_{OD}	1.1910	1.2962	0.9601	1.4227	1.3818	1.3699	1.3930	1.2869	**1.4519**
San Diego	$AUC_{(PD,PF)}$	0.8885	0.9717	0.8849	0.9836	0.8730	0.9785	0.9913	0.9939	0.9957
	$AUC_{(PD,\tau)}$	0.0686	0.0609	0.1584	0.3171	0.1248	0.0555	0.2403	0.6023	0.4754
	$AUC_{(PF,\tau)}$	0.0381	0.0129	0.0805	0.0591	0.0619	0.0107	0.0066	0.0779	0.0775
	AUC_{OD}	0.9190	1.0197	0.9628	1.2416	0.9359	1.0233	1.2250	**1.5183**	1.3936
Airport-1	$AUC_{(PD,PF)}$	0.8221	0.7489	0.8023	0.8782	0.7750	0.8045	0.9149	0.9161	0.9475
	$AUC_{(PD,\tau)}$	0.0987	0.1139	0.1825	0.2764	0.2347	0.0559	0.1345	0.3416	0.3064
	$AUC_{(PF,\tau)}$	0.0424	0.0718	0.1127	0.1296	0.1480	0.0292	0.0370	0.1057	0.0873
	AUC_{OD}	0.8784	0.7910	0.8721	1.0250	0.8617	0.8313	1.0123	1.1521	**1.1665**
Airport-2	$AUC_{(PD,PF)}$	0.8404	0.9027	0.8891	0.8858	0.8344	0.9411	0.9735	0.9485	0.9808
	$AUC_{(PD,\tau)}$	0.1841	0.1137	0.2452	0.3595	0.2204	0.1691	0.3083	0.2780	0.4154
	$AUC_{(PF,\tau)}$	0.0516	0.0424	0.0982	0.1467	0.1151	0.0331	0.0278	0.1088	0.0880
	AUC_{OD}	0.9729	0.9741	1.0361	1.0986	0.9398	1.0771	1.2539	1.1177	**1.3083**
Airport-3	$AUC_{(PD,PF)}$	0.9288	0.8817	0.9297	0.9349	0.9073	0.9370	0.9522	0.9526	0.9474
	$AUC_{(PD,\tau)}$	0.0660	0.0452	0.2211	0.2922	0.0491	0.0262	0.0810	0.1906	0.1883
	$AUC_{(PF,\tau)}$	0.0145	0.0079	0.0504	0.0802	0.0107	0.0019	0.0078	0.0465	0.0471
	AUC_{OD}	0.9804	0.9191	1.1004	**1.1469**	0.9457	0.9613	1.0253	1.0967	1.0886
Airport-4	$AUC_{(PD,PF)}$	0.9526	0.9845	0.9432	0.9795	0.9785	0.9748	0.9934	0.9911	0.9911
	$AUC_{(PD,\tau)}$	0.0736	0.3527	0.2981	0.4289	0.4725	0.2385	0.4357	0.6356	0.5330
	$AUC_{(PF,\tau)}$	0.0248	0.0609	0.0789	0.0972	0.1359	0.0344	0.0806	0.0777	0.1097
	AUC_{OD}	1.0015	1.2764	1.1624	1.3113	1.3150	1.1789	1.3486	**1.5490**	1.4144
Beach-1	$AUC_{(PD,PF)}$	0.9807	0.9769	0.9890	0.9779	0.9277	0.9741	0.9752	0.9776	0.9848
	$AUC_{(PD,\tau)}$	0.2496	0.2637	0.4710	0.2877	0.2467	0.1353	0.2642	0.2776	0.3376
	$AUC_{(PF,\tau)}$	0.0065	0.0058	0.0386	0.0208	0.0510	0.0026	0.0255	0.0158	0.0174
	AUC_{OD}	1.2238	1.2347	**1.4213**	1.2448	1.1234	1.1068	1.2139	1.2394	1.3051
Beach-2	$AUC_{(PD,PF)}$	0.9106	0.9097	0.8061	0.9278	0.8603	0.9031	0.9411	0.9011	0.9571
	$AUC_{(PD,\tau)}$	0.1530	0.1124	0.1772	0.2187	0.1622	0.1069	0.1931	0.1114	0.2917
	$AUC_{(PF,\tau)}$	0.0488	0.0324	0.0901	0.0463	0.0728	0.0285	0.0318	0.0545	0.0595
	AUC_{OD}	1.0148	0.9897	0.8932	1.1002	0.9497	0.9814	1.1024	0.9580	**1.1892**
Beach-3	$AUC_{(PD,PF)}$	0.9998	0.9995	0.9982	0.9934	0.9988	0.9996	0.9995	0.9941	0.9998
	$AUC_{(PD,\tau)}$	0.5314	0.5318	0.6149	0.5475	0.6817	0.5367	0.6067	0.5421	0.7499
	$AUC_{(PF,\tau)}$	0.0259	0.0172	0.0744	0.0963	0.0951	0.0241	0.0474	0.0581	0.0527
	AUC_{OD}	1.5053	1.5142	1.5386	1.4446	1.5853	1.5122	1.5589	1.4781	**1.6969**
Beach-4	$AUC_{(PD,PF)}$	0.9538	0.9603	0.9583	0.9820	0.9595	0.9690	0.9739	0.9691	0.9669
	$AUC_{(PD,\tau)}$	0.1343	0.0878	0.3820	0.2448	0.1699	0.1341	0.2094	0.2678	0.3558
	$AUC_{(PF,\tau)}$	0.0233	0.0061	0.0774	0.0226	0.0329	0.0115	0.0243	0.0342	0.0555
	AUC_{OD}	1.0648	1.0421	1.2629	1.2043	1.0965	1.0916	1.1589	1.2027	**1.2671**
Urban-1	$AUC_{(PD,PF)}$	0.9907	0.9922	0.9390	0.8491	0.9598	0.9832	0.9914	0.9828	0.9956
	$AUC_{(PD,\tau)}$	0.3143	0.2705	0.4413	0.4031	0.4226	0.3994	0.4898	0.5625	0.5618
	$AUC_{(PF,\tau)}$	0.0556	0.0367	0.1522	0.2073	0.1442	0.0558	0.0790	0.1041	0.1126
	AUC_{OD}	1.2494	1.2260	1.2282	1.0449	1.2382	1.3269	1.4023	1.4412	**1.4447**
Urban-2	$AUC_{(PD,PF)}$	0.9946	0.9957	0.9409	0.8761	0.8766	0.9839	0.9638	0.9950	0.9980
	$AUC_{(PD,\tau)}$	0.1178	0.1446	0.3776	0.1424	0.0650	0.0568	0.2610	0.2280	0.2601
	$AUC_{(PF,\tau)}$	0.0135	0.0138	0.1032	0.0398	0.0181	0.0038	0.0747	0.0327	0.0475
	AUC_{OD}	1.0989	1.1265	**1.2153**	0.9788	0.9235	1.0368	1.1501	1.1903	1.2106
Urban-3	$AUC_{(PD,PF)}$	0.9513	0.9577	0.8224	0.9420	0.9512	0.9634	0.9799	0.9317	0.9899
	$AUC_{(PD,\tau)}$	0.0963	0.1079	0.1604	0.4619	0.3889	0.1471	0.3816	0.3667	0.4329
	$AUC_{(PF,\tau)}$	0.0351	0.0327	0.0729	0.1231	0.1187	0.0306	0.0510	0.1124	0.0667
	AUC_{OD}	1.0125	1.0329	0.9100	1.2808	1.2214	1.0798	1.3105	1.186	**1.3561**
Urban-4	$AUC_{(PD,PF)}$	0.9887	0.9871	0.9617	0.9350	0.7192	0.9802	0.9910	0.9961	0.9897
	$AUC_{(PD,\tau)}$	0.0891	0.1034	0.1644	0.0784	0.0281	0.0409	0.1515	0.1305	0.1187
	$AUC_{(PF,\tau)}$	0.0114	0.0115	0.0322	0.0102	0.0012	0.0017	0.0292	0.0114	0.0158
	AUC_{OD}	1.0664	1.0791	1.0940	1.0032	0.7461	1.0195	1.1134	**1.1151**	1.0926
Urban-5	$AUC_{(PD,PF)}$	0.9692	0.9658	0.9370	0.9325	0.9086	0.9612	0.9574	0.9630	0.9749
	$AUC_{(PD,\tau)}$	0.1461	0.1507	0.3526	0.2787	0.2271	0.1477	0.2741	0.4085	0.3887
	$AUC_{(PF,\tau)}$	0.0437	0.0453	0.1219	0.0903	0.0920	0.0342	0.0710	0.0713	0.1012
	AUC_{OD}	1.0716	1.0712	1.1677	1.1210	1.0436	1.0747	1.1605	**1.3003**	1.2624

Figs. 7–10, we separately present the influence of different parameters on San Diego and Airport-1 datasets. Each column represents the results relating to a different p. With the increase of p, the value of $AUC_{PD,PF}$ (hereafter abbreviated as AUC) diminishes gradually when α approaches zero, which means that a smaller p gives a better detection result. For

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TABLE II
REFERENCE FOR PARAMETERS IN S^{2}ELR FOR EACH DATASET

HSI datasets	Parameters(r,α,β,p)	HSI datasets	Parameters(r,α,β,p)
HYDICE_urban	(7,0.01,0.013,0.4)	Beach-3	(9,0.03,0.08,0.1)
San Diego	(11,0.01,0.5,0.1)	Beach-4	(3,0.01,0.02,0.4)
Airport-1	(3,0.027,0.7,0.1)	Urban-1	(11,0.01,0.3,0.3)
Airport-2	(7,0.01,0.03,0.1)	Urban-2	(5,0.01,0.04,0.9)
Airport-3	(3,0.01,0.02,0.1)	Urban-3	(5,0.01,0.04,0.2)
Airport-4	(11,0.01,0.03,0.7)	Urban-4	(11,0.02,0.07,0.1)
Beach-1	(11,0.01,0.07,0.1)	Urban-5	(5,0.01,0.09,0.6)
Beach-2	(9,0.01,0.01,0.1)		

TABLE III
COMPARISON OF AUC VALUES FOR ABLATION EXPERIMENTS

Datasets	Without Laplacian		Unweighted		Without PCA		$l_{2,2,1}$ norm		S^{2}ELR	
	AUC	Time	AUC	Time	AUC	Time	AUC	Time	AUC	Time
HYDICE_urban	0.9855	479.56	0.9937	454.82	0.9771	7794.52	0.9883	18.60	0.9946	256.14
Sandiego	0.9952	195.38	0.9954	208.04	0.9985	9888.46	0.9933	6.05	0.9959	119.68
Airport-1	0.9336	114.67	0.9378	159.69	0.9346	12350.62	0.6748	22.74	0.9475	68.39
Airport-2	0.9616	163.99	0.9644	167.65	0.8061	11232.47	0.9077	33.40	0.9808	99.26
Airport-3	0.9420	663.59	0.9505	744.87	0.7749	10113.66	0.9281	18.78	0.9560	485.99
Airport-4	0.9878	705.26	0.9933	596.49	0.9354	11313.33	0.9860	22.95	0.9933	514.88
Average	0.9676	387.07	0.9725	388.59	0.9044	10448.84	0.9130	20.42	0.9780	257.39

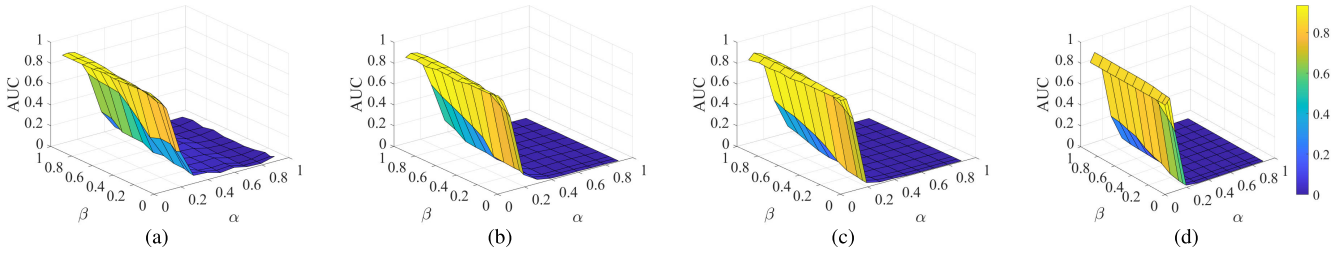


Fig. 10. AUC values for Airport-1 dataset under various p ($r = 9$). (a) $p = 0.1$. (b) $p = 0.3$. (c) $p = 0.5$. (d) $p = 0.7$.

each row, it denotes the results corresponding to a different r. Limited by the space constraints, we only give results for $r = 5$ and $r = 7$ on both datasets, and as can be seen by comparing each row in the figures, our S^{2}ELR method is robust to r and β. Moreover, Table II lists the optimal parameters corresponding to all datasets as references.

For academic exchanges and fair comparison, we give the parameters setting of all competing methods by following the original papers and [6], [11], and [28]. Specifically, RX and TPCA are parameter-free. The parameters of RPCA are set to code defaults ($\lambda = 0.01$ or 0.001). Then, the parameters of GTVLRR are fine-tuned by referring to the values given in [20] ($\lambda = 0.5$, $\beta = 0.2$, $\gamma = 0.05$, $k = 10$ and $\sigma = 1$). For LRASR, we choose β and λ from $[0.001, 0.01, 0.1, 0.5, 1, 2, 3]$ and $[0.001, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5]$, respectively. And for LSMAD, we take the same empirical value of 1-2 for rank as in [14]. For DeCNN-AD method, the parameters λ and β are both taken in the range of 0.0001–0.3. For PTA, the rank takes values from 1 to 20 and the hyperparameters (α, β, τ) take values from 0.01 to 1.

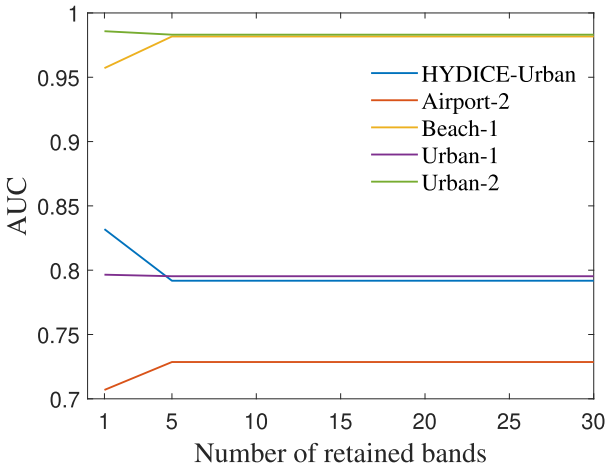


Fig. 11. Effect of retained bands on anomaly detection results.

2) Effect of Laplace Regularization on Results: To better exploit the local structure details and augment the correlation of background, we introduced the Laplacian regularizer to guarantee the smoothness of the background. As listed in the

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TABLE IV
RUNNING TIME (S) OF ALL COMPARED METHODS

HSI Data sets	RX	RPCA	TPCA	GTVLRR	LRASR	LSMAD	DeCNN-AD	PTA	S^{2}ELR
HYDICE-Urban	0.065	1.443	6.588	166.595	15.278	3.438	66.181	10.303	256.143
San Diego	0.030	1.619	16.749	63.566	15.771	5.507	14.044	12.901	119.681
Airport-1	0.034	13.904	24.620	80.503	16.598	6.395	72.965	14.056	68.394
Airport-2	0.032	3.966	21.855	156.465	17.944	6.024	18.553	14.055	99.262
Airport-3	0.040	4.278	16.662	221.604	18.450	5.690	14.154	14.154	485.991
Airport-4	0.032	1.739	16.840	233.939	16.377	5.365	38.986	13.438	514.883
Average	0.039	4.492	17.219	153.779	16.736	5.403	37.481	13.151	257.392

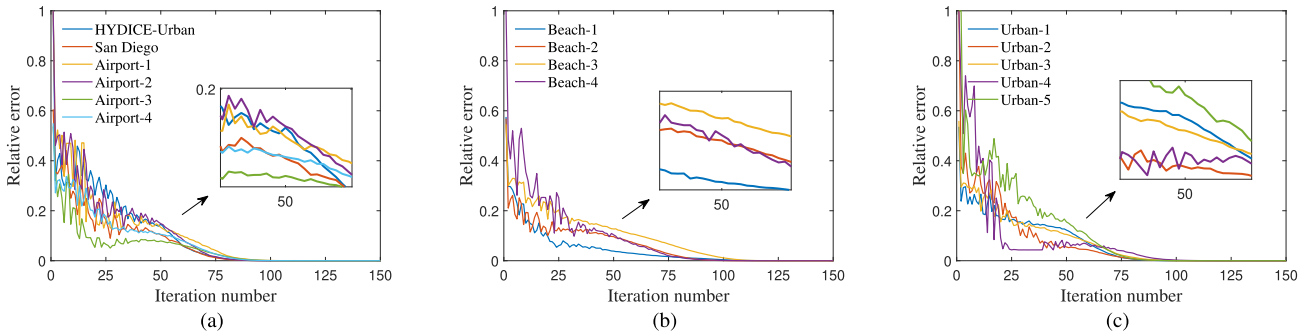


Fig. 12. Relative errors on all tested datasets. (a) HYDICE-Urban, San Diego, and Airport 1-4 datasets. (b) Beach 1-4 datasets. (c) Urban 1-5 datasets.

first column of Table III, the AUC metrics of S^{2}ELR with the Laplacian regularizer are higher than those without this term. Specifically, in the HYDICE-Urban dataset, the S^{2}ELR method with the Laplacian regularizer is nearly 0.01 higher in comparison to the AUC value with no such term, while in the four Airport datasets, these are 0.0138, 0.0187, 0.014, and 0.0055 higher, respectively. Consequently, it is quite evident that the Laplace regular term carries a tremendously significant role in improving AD performance.

3) Effect of Weighted and Unweighted Tensor Schatten-p Norm: Following WTNN [48], we introduce the weighted Schatten-p norm to diminish the contribution of large singular values on the rank() function. To illustrate the impact of the weights, we design a comparative experiment and conclude the results in the second column of Table III. Apparently, since the weighted coefficients balance the contribution of singular values in different sizes to the rank() function, the results for the weighted tensor Schatten-p norm are always better than those without weighted. For instance, the AUC value for the Airport-1 dataset with weighted is nearly 0.02 higher than that unweighted, and the overall average value is almost 0.006 higher. It is further demonstrate that the weighted tensor Schatten-p norm is a tighter approximation of the tensor rank.

4) Effect of PCA: As mentioned before, to alleviate the computation burden, we have applied PCA to preprocess the original data by degrading its dimension. Subsequently, to illustrate the effect of PCA, the third column of Table III presented the results comparison of data without PCA under the same parameters. First, the most noticeable difference in Table III is the time. Since the original data has been drastically degraded in scale, the average running time of our S^{2}ELR with PCA is astonishingly shortened by almost 40 times. The second is the AUC value. As shown in Table III, the average AUC value overall was 7% higher than that without PCA, except for a relatively lower AUC value on San Diego dataset with PCA. Actually, thanks to the contribution of PCA, most of the redundant background pixels are eliminated, which allows the anomalies to be more easily separated and therefore the results perform better. Besides this, we randomly present the effect of different retained bands on anomaly detection for several datasets in Fig. 11. As can be seen from Fig. 11, there is no significant change in the results of anomaly detection with the increase of retained bands, whereas the computing cost grows exponentially. Thus, it also demonstrates the effectiveness and superiority of the PCA method from another side.

5) Effect of Structured Sparsity-Inducing Norm: To further elucidate the superiority of the structured sparsity-inducing norm, we have compared it with the routine $l_{2,2,1}$ norm, which depicts the sparsity of tensors, and the results are summarized in the penultimate column of Table III. It can be observed that the structured sparsity-inducing norm achieves all better results compared to the $l_{2,2,1}$ norm due to the adequate exploration of the sparsity between each slice along the spatial dimensions. Specifically, our S^{2}ELR is nearly 30% higher in the Airport-1 dataset, while the average is 6.5% higher.

D. Complexity Analysis and Running Time

In this section, we first present the complexity analysis of our S^{2}ELR method. By referring to Algorithms 1 and 2, for an HSI of size $n_1 \times n_2 \times b$ ($n_1 \geqslant n_2$), the primary computing costs of the proposed method are as follows.

1) *Update $\mathcal{B}$:* To begin with, the complexity of fft and its inverse ifft is $\mathcal{O}(n_1 n_2 b \times \log(n_1 n_2 b))$. Then, the complexity of computing SVD for each frontal slice

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is $\mathcal{O}(n_1^2 n_2)$. After that, the complexity of the two-level *for* loop is $\mathcal{O}(n_2 \times b)$. Finally, the total complexity of updating $\mathcal{B}$ is $\mathcal{O}(n_1 n_2 b \times \log(n_1 n_2 b) + n_1^2 n_2 \times b + n_2 b)$.

2) *Update* $\mathcal{E}$*:* According to (28), the complexity of the proximal operator used to solve $\mathcal{E}$ is $\mathcal{O}(n_1 n_2 b)$.
3) *Update* $\mathcal{Z}$*:* According to (31), the complexity of matrix multiplication and inversion are $\mathcal{O}(n_1^2 n_2^2 b)$ and $\mathcal{O}(n_1^3 n_2^3)$, respectively. Furthermore, the complexity of matrix addition or subtraction is $\mathcal{O}(n_1 n_2 b)$. Hence, the practical complexity of solving $\mathcal{Z}$ is $\mathcal{O}(n_1^2 n_2^2 b + n_1^3 n_2^3 + n_1 n_2 b)$.

Therefore, the whole complexity of S^{2}ELLR is $\mathcal{O}(n_1^2 n_2^2 b + n_1^3 n_2^3 + n_1^2 n_2 b + n_1 n_2 b)$.

The running times of all compared methods are listed in Table IV. Obviously, it can be observed that since most LRR-based methods inevitably utilize the SVD, such as GTVLRR, LRASR, and the proposed S^{2}ELR methods suffer from the high running time. Besides, the DNN-based method (DeCNN-AD) is also unexpectedly time consuming. To some extent, it is worthwhile to sacrifice some running time for improving detection performance.

E. Convergence Analysis

The relative error curves trend for all 15 tested datasets is shown in Fig. 12. It can be seen that although there exist some fluctuations during the first 25 iterations, they finally converge to zero after at most 50 iterations.

V. Conclusion

In this article, we proposed a novel S^{2}ELR method for HAD by integrating the weighted tensor Schatten-p norm, the position-based Laplacian regularizer and the structured sparsity-inducing norm. Specifically, the first two terms were exploited to uncover the correlations among each frontal slice and the local smoothing of the background pixels. Besides, the structured sparsity-inducing norm was adopted to detect the anomalies, which incorporated the abnormal pixels within each band on the spectral dimension. Finally, the ADMM was introduced to solve the S^{2}ELR optimization problem. In comparison with several existing state-of-the-art methods, experimental results on 15 public datasets also demonstrated the superiority of our method.

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